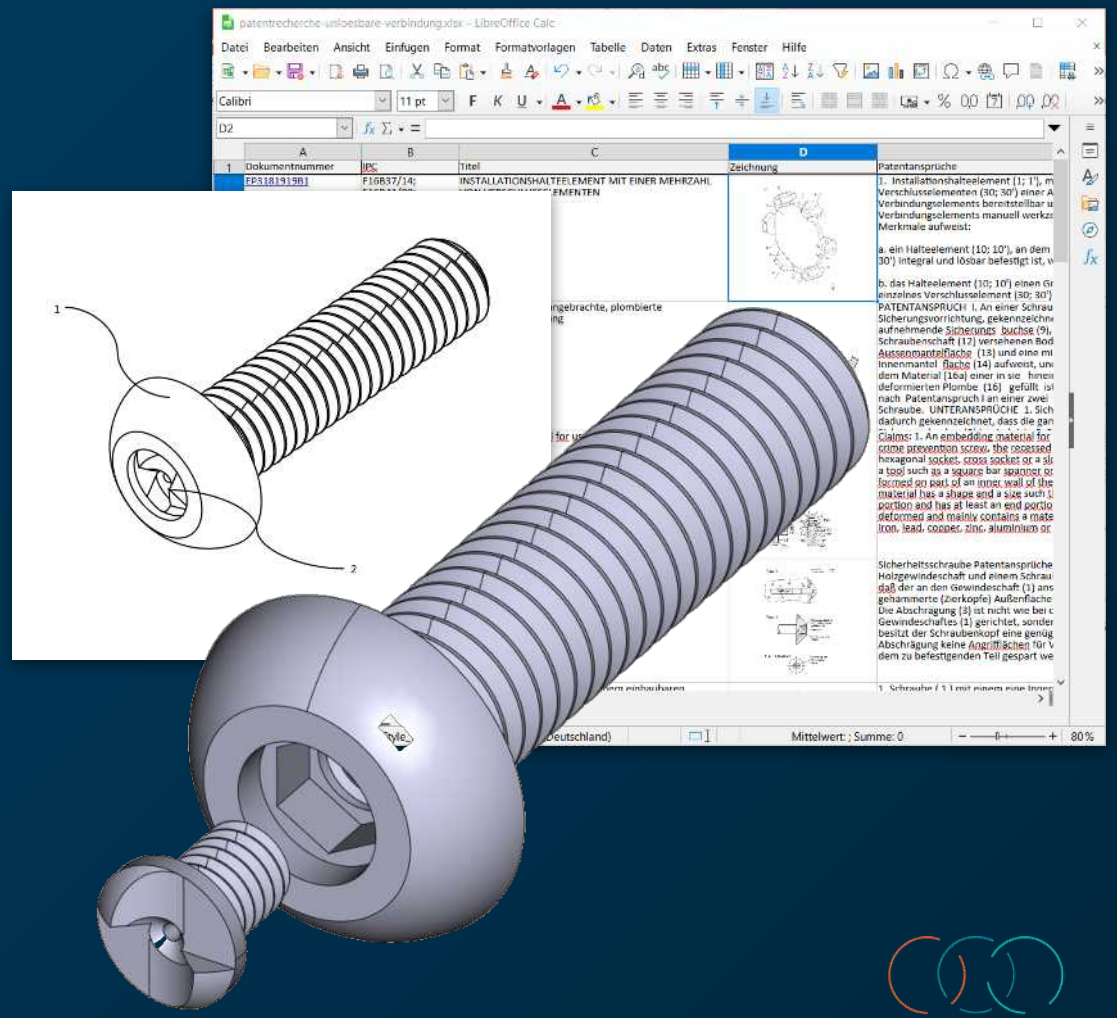
ALSADO

Keynote

Inventor’s Handbook

Guide to the product development process using open-source software

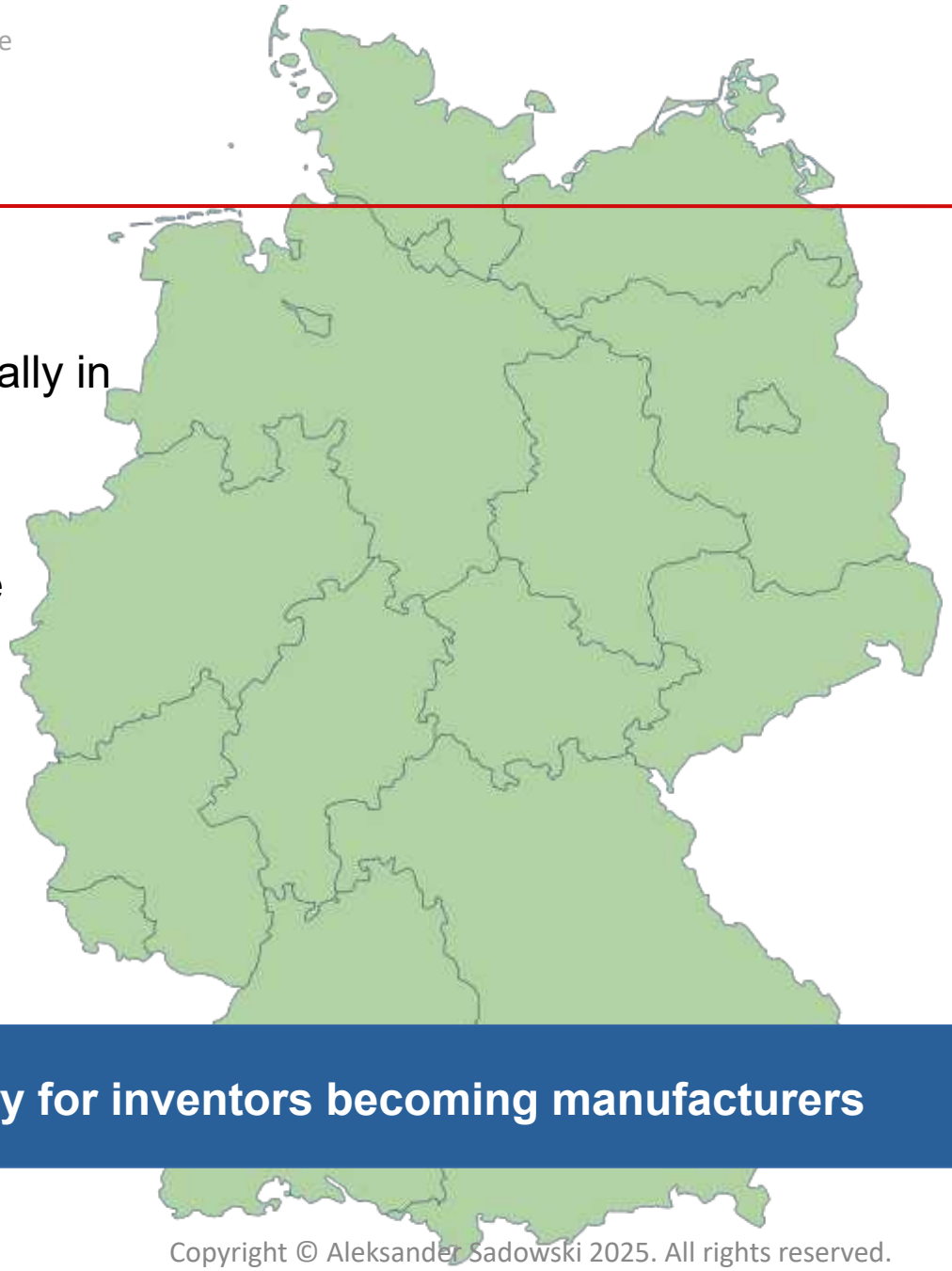


Aleksander Sadowski

OPENCHAIN

Manufacturing in Germany

- Germanys industry is on an downwards trends
- Innovation and Trust are the biggest assets (especially in automotive and mechanical engineering)
- Boost inventions and their implementation
- Entrepreneurs need lower barrier to entry and more stability



=> Open-source software lowers the barrier to entry for inventors becoming manufacturers

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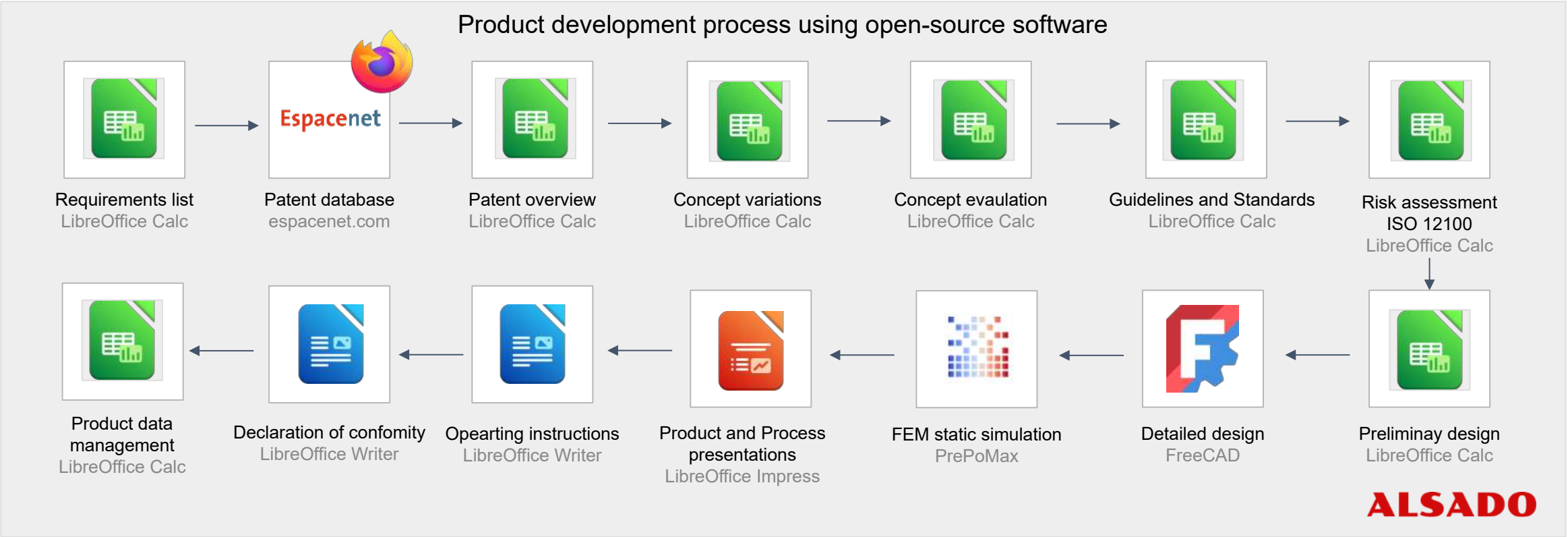
Overview of the Product development process using open-source software



=> Product development can be fully implemented using open-source software

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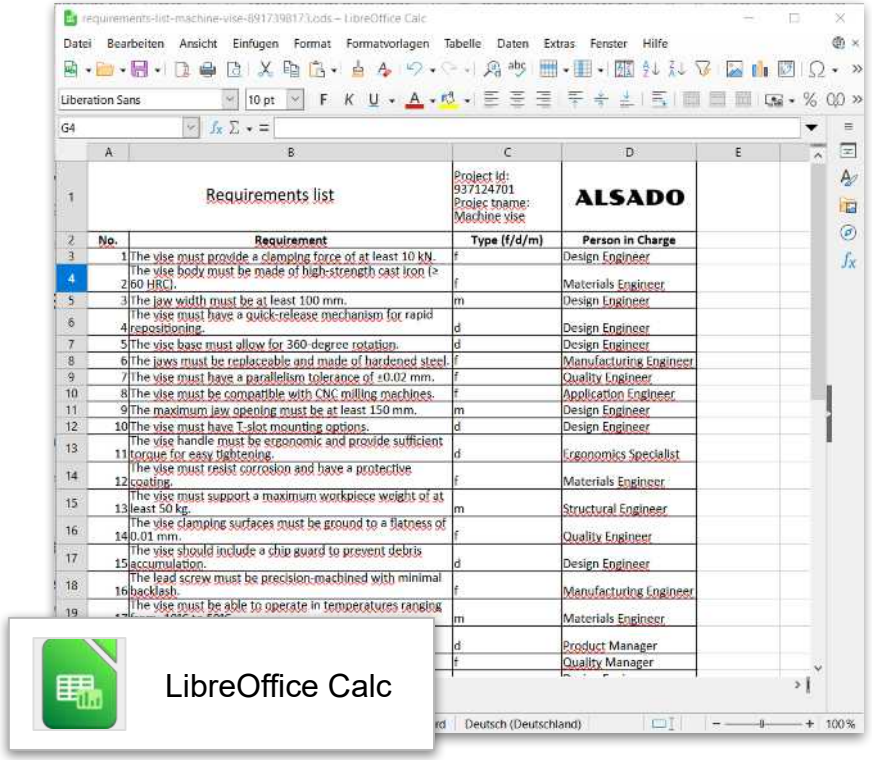
Defining the product requirements

- Clearly defining product requirements based on from market research
- Including findings from analysis on existing product and its functional structure
- Requirements as rows with:
  - „Nr.“
  - „Anforderung“
  - „Typ (f/d/m)“
  - „Verantwortliche Person“



=> Requirements list keeps all project members on track for a common goal

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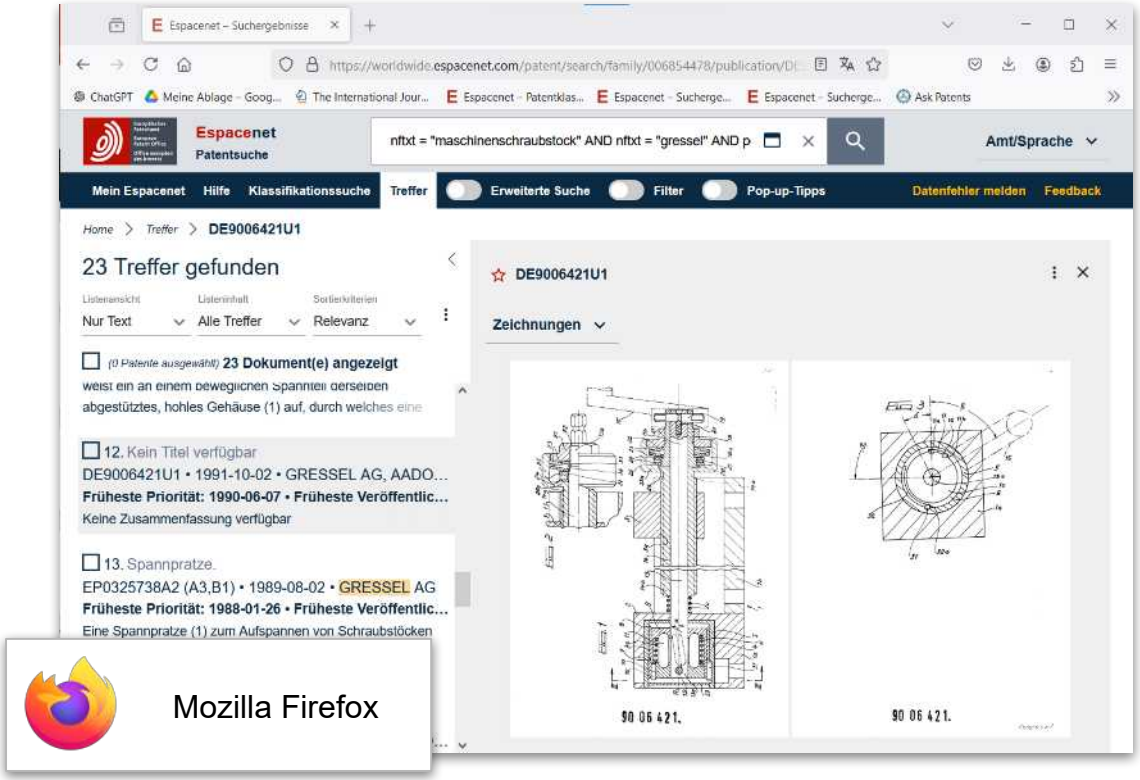
Finding technologies in public domain through a patent search

- Effectively reducing **160 million** patent documents to **100** and finding suitable solutions for our product.
- Finding documents, that are 20+ years old to prevent patent infringements
- Finding most suitable IPC/ CPC
- Using keywords from the requirements list as filters for the query



=> Utilizing inventions in public domain as a legally safe basis for further developments

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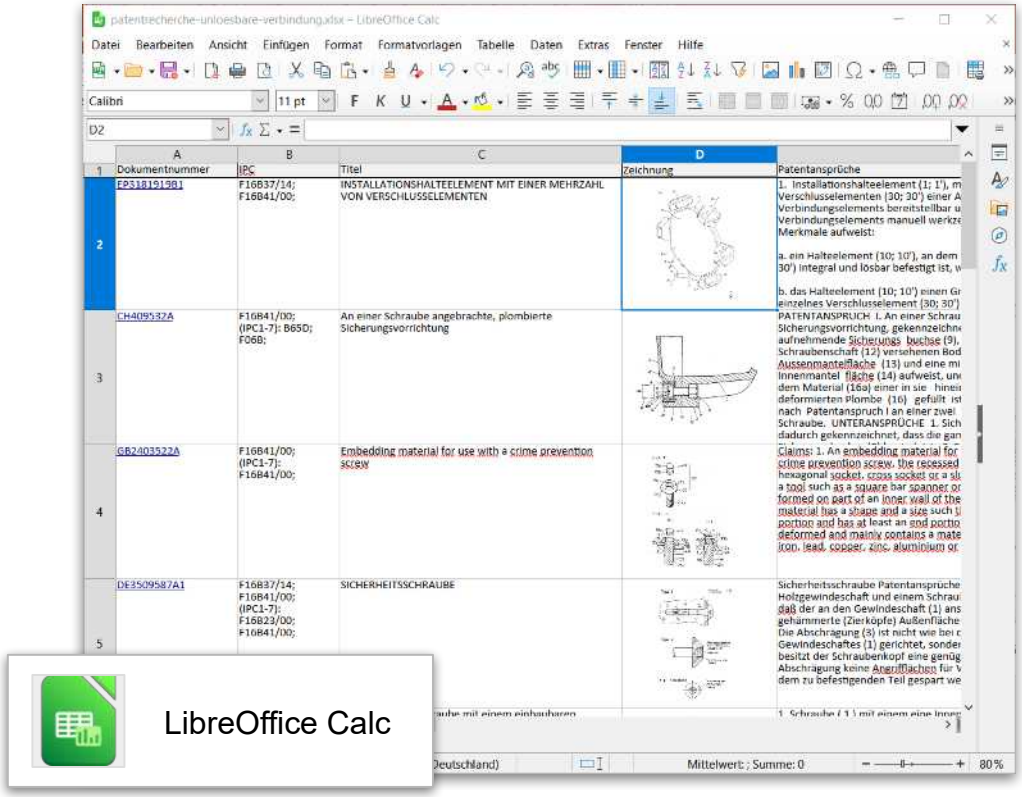
Managing suitable solutions from the patent search

- Keep an overview of your verified solutions
- Identify patterns in the technology and find gaps to predict future innovations
- Extending with solutions from other sources (catalogues, books, …)



=> Preventing patent infringements by managing verified solutions

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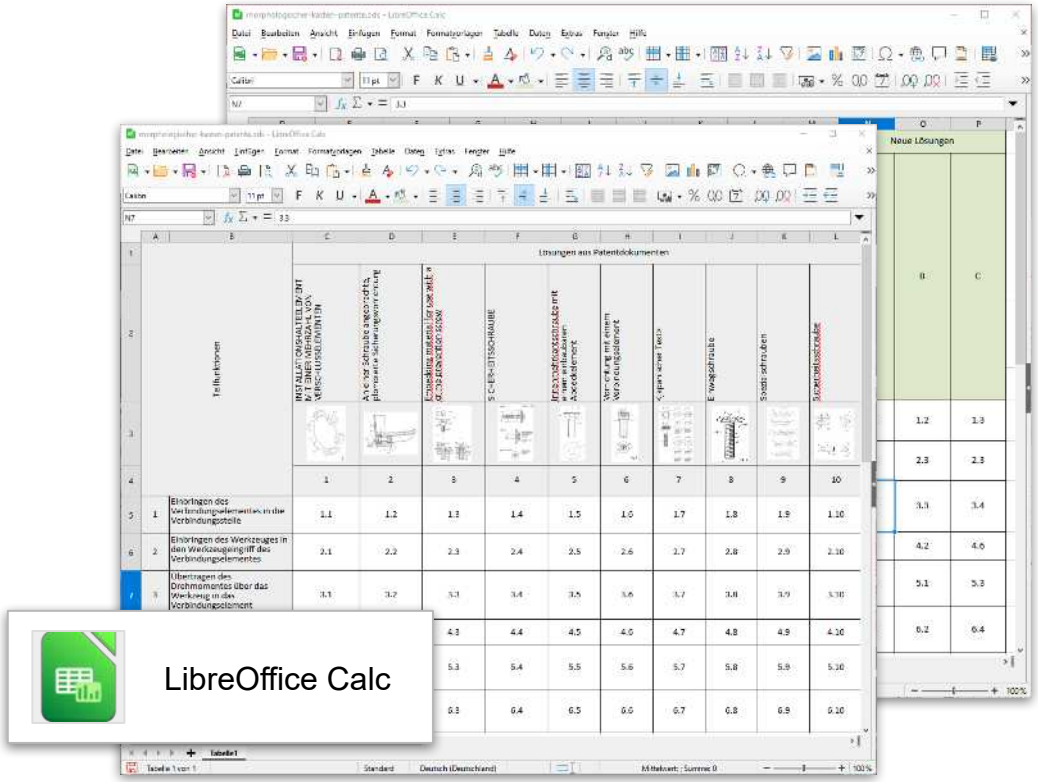
Morphological box for product sub-functions

- Defining a matrix with with pairs of potential solutions for each sub-function and verified patent as 1.1, 1.2, …
- Sub-functions as rows, patents as columns
- Creating a complete product solutions, by picking different combinations of patents for each sub-function



see also: Roloff/ Matek Maschinenelemente

=> Keeping solutions open and finding multiple complete solutions for a product

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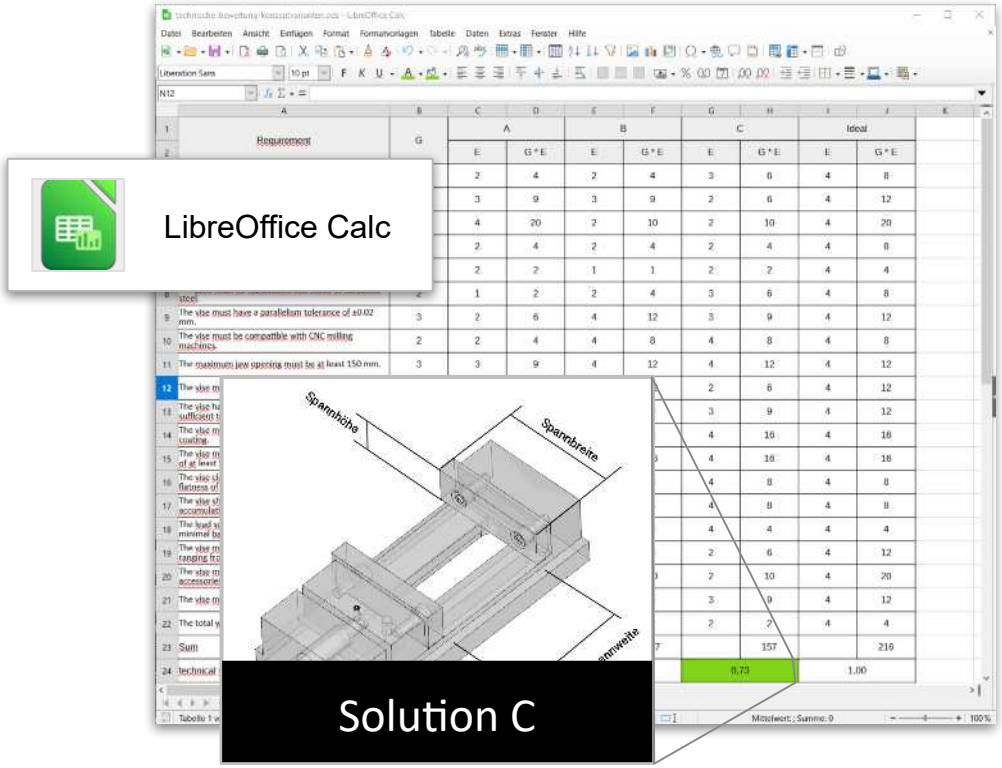
Evaluating and comparing different product solutions

- Comparing multiple product solutions using a systematic approach
- Evaluating each solutions with respect to the product requirements and their individual importance
- Creating a product concept as a sketch



see also: Roloff/ Matek Maschinenelemente

=> Systematically finding the best solution for the new product

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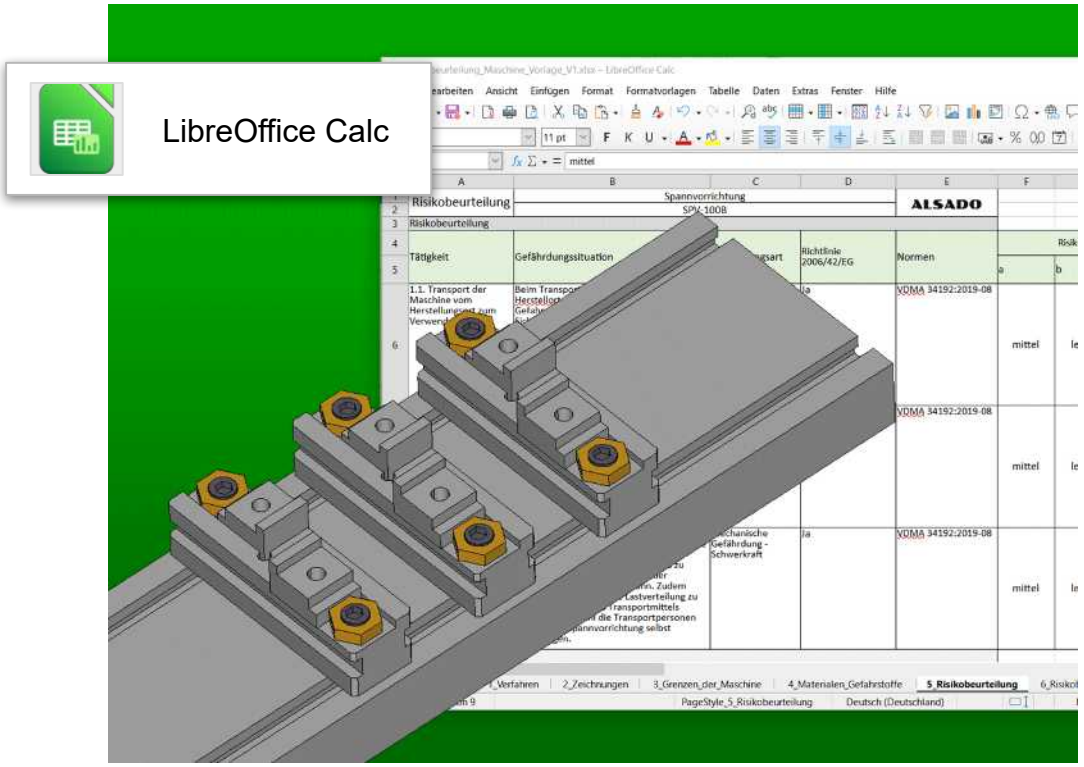
Product safety

- Getting an overview of all available (EU) guidelines and standards
- Finding the guidelines and standards that apply for the product
- Creating a risk assessment ISO 12100
- Often overlooked by inventors and very rarely mentioned in universities



=> Building safe products prevents future liability and builds trust

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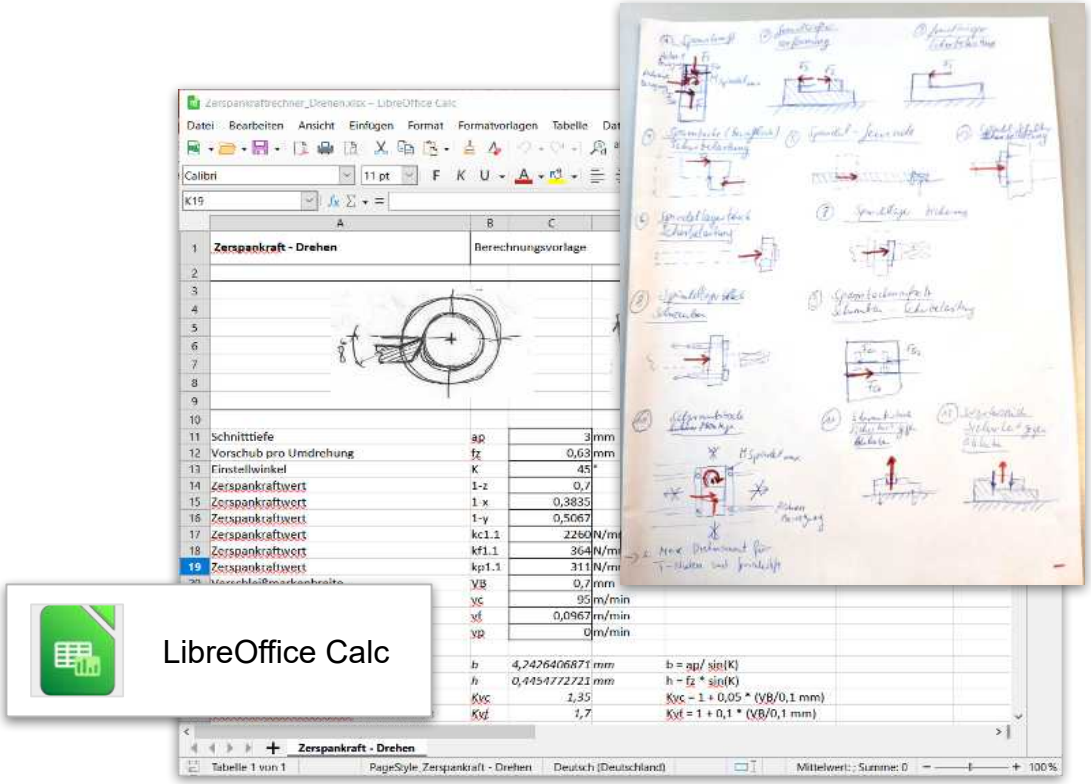
Preliminary design

- Determining product properties based on product concept and product safety:
  - safety factor
  - forces
  - dimensions
  - tolerances
  - surfaces
  - temperature
  - …



=> Determine product properties to utilize those in detailed design

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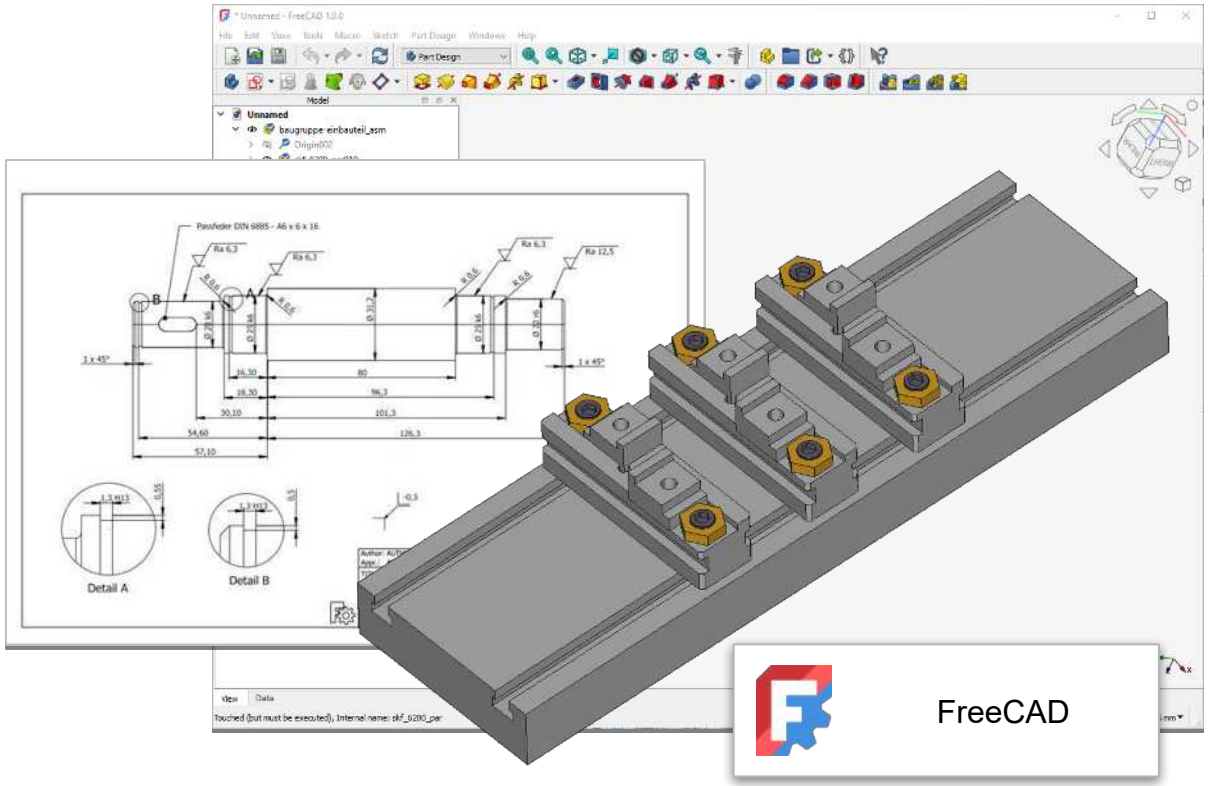
Detailed design in FreeCAD

- 3D-CAD-software with universal Interface: Import und Export STEP, DXF, PDF
- Modular framework:
  - Part Design (parts)
  - A2Plus (assemblies)
  - TechDraw (technical drawings)
  - FEM (simple FEM simulations)
- Python-API: Intuitively understand scripting and automate tasks
- Community: Wiki, Forum and social media



=> FreeCAD is a complete 3D CAD software

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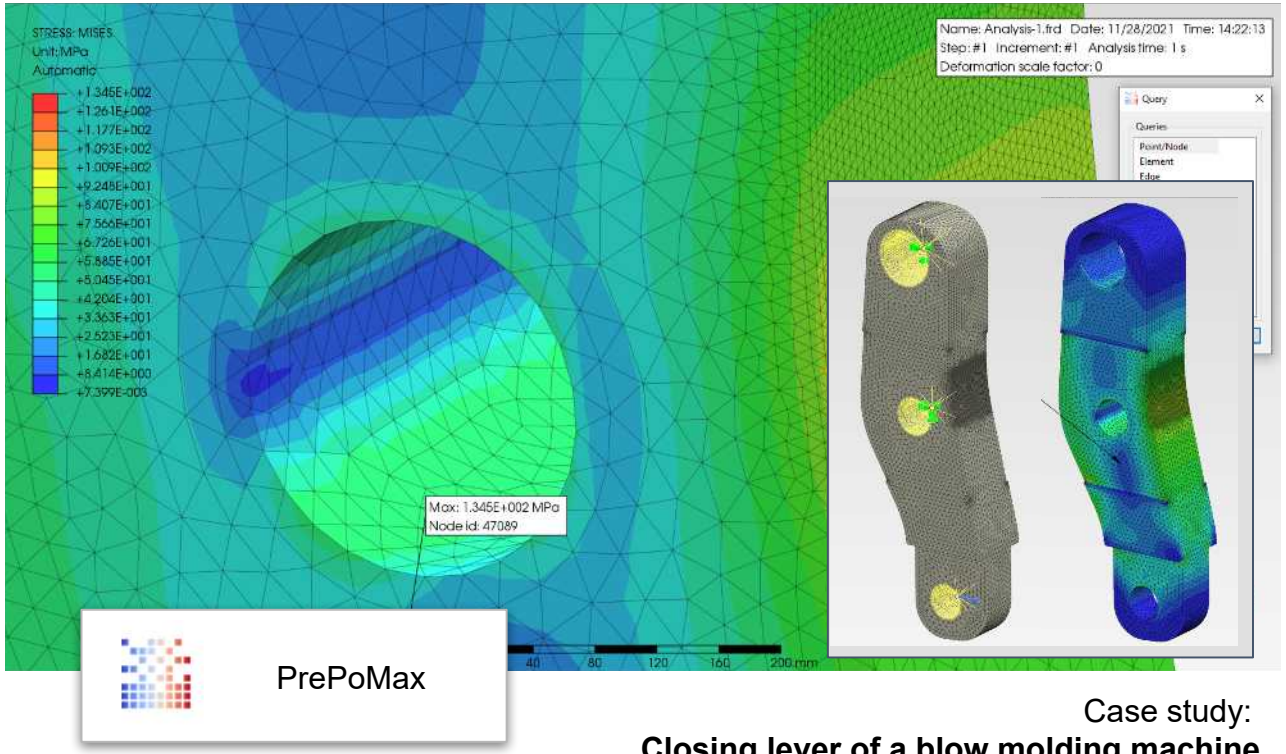
Simulation with Finite-Element-Method

- PrePoMax GUI and workflow similar to Abaqus
- PrePoMax with Calculix solver for static analysis
- PrePoMax with OpenRadioss solver for dynamic analysis



Case study:
Closing lever of a blow molding machine

=> PrePoMax is similar to Abaqus and can be used as a pre-prcessor for OpenRadioss

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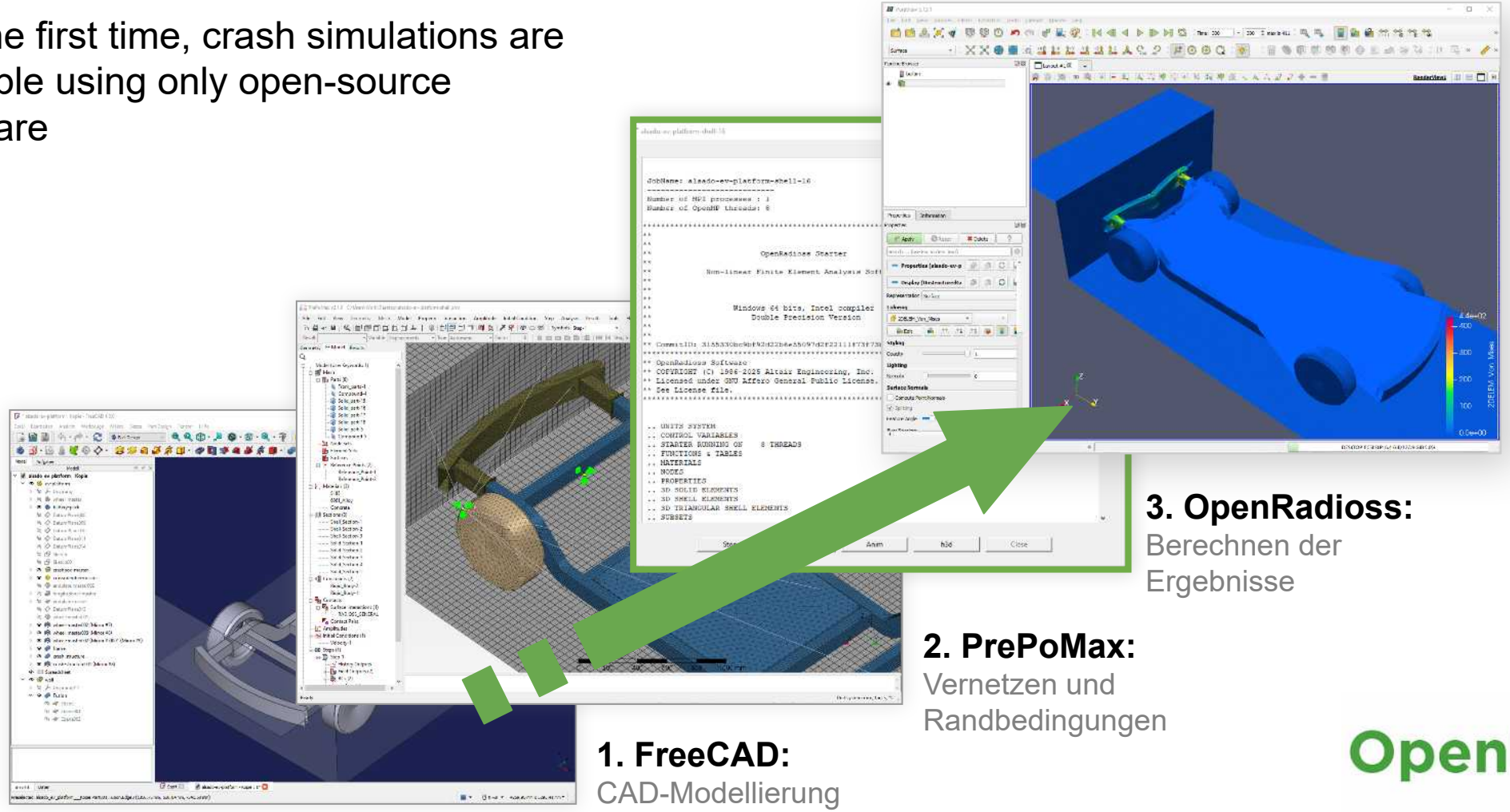
Dynamic simulations with open-source software

For the first time, crash simulations are possible using only open-source software



1. FreeCAD:
CAD-Modellierung

2. PrePoMax:
Vernetzen und Randbedingungen

3. OpenRadioss:
Berechnen der Ergebnisse

4. ParaView:
Visualisierung der Ergebnisse

OpenRadioss™

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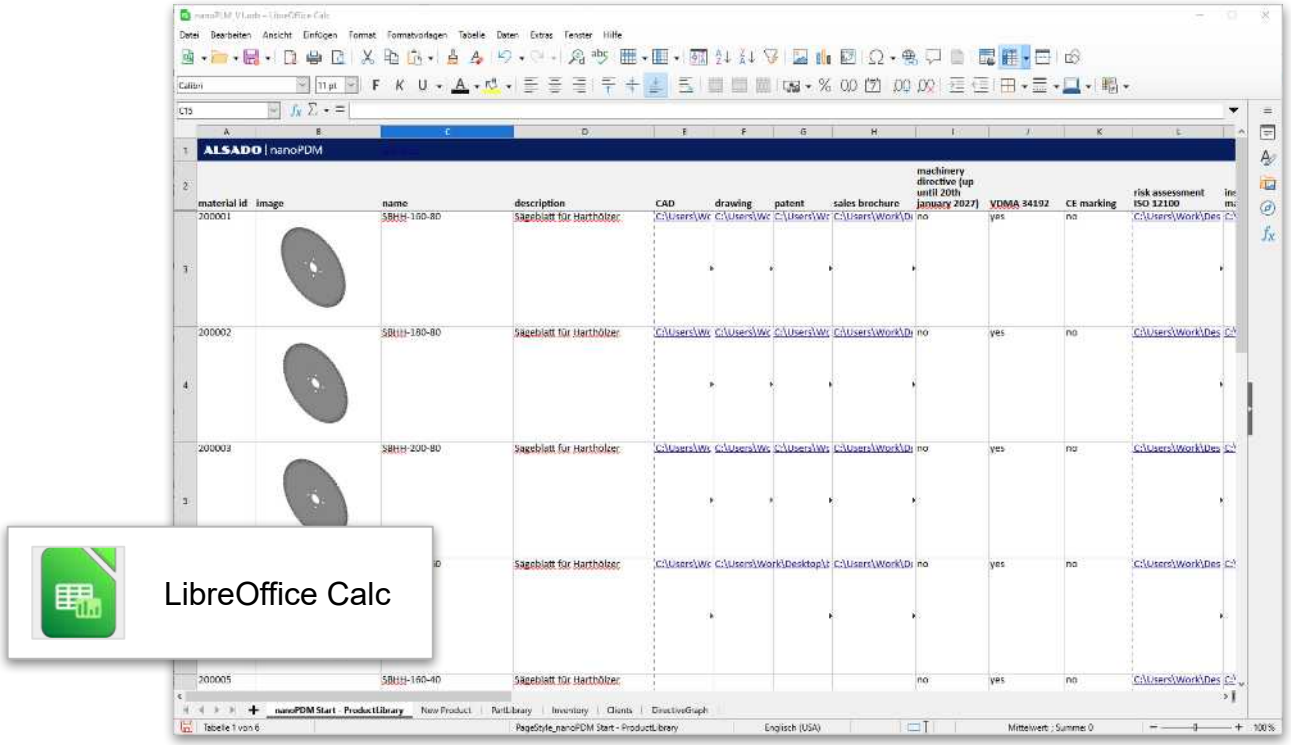
Product data management using simple tools

- Manage all your parts and assemblies in a centralized spreadsheet with all associated files (drawings, legal documents, patents)
- Easily accessible for non-engineers
- Assign clients to serial numbers



=> Getting an overview of all your parts, products and associated files

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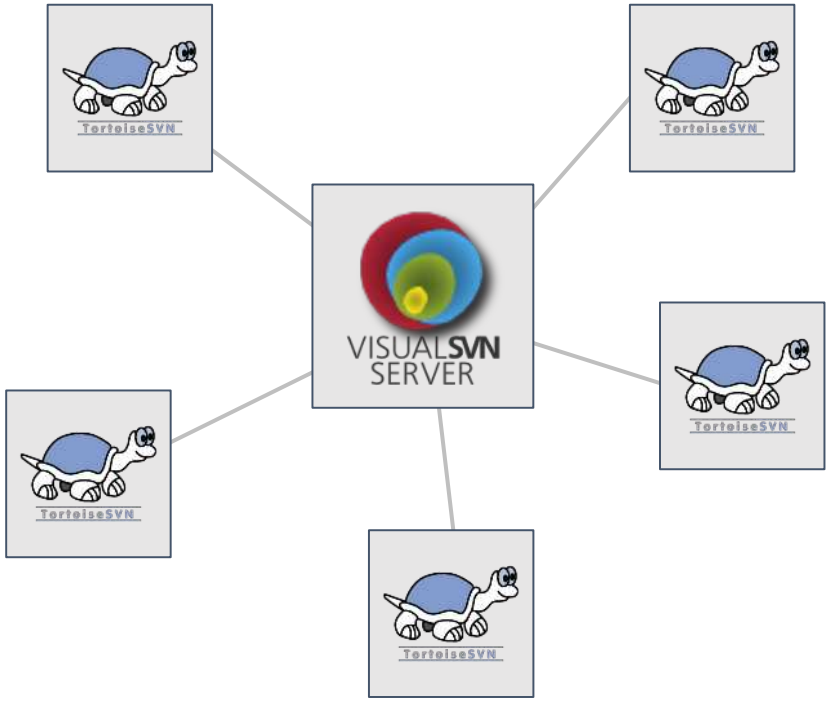
Create an infrastructure for your team

- Setup versioning system for product data without programming skills (VisualSVN + TortoiseSVN)
- Let your team collaborate and manage their roles
- Prevent file duplicates
- Centralized server in your organsation with redundancy and a battery backup
- Back up data in regular time intervals and store it in a different location



=> Secure your product data and enable collaboratin in a team

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Open-source-suite for product development in mechanical engineering

Open-source-suite for product development in mechanical engineering

Mozilla Firefox	LibreOffice Calc	LibreOffice Writer	LibreOffice Impress	FreeCAD	PrePoMax

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=> We offer support for open-source software

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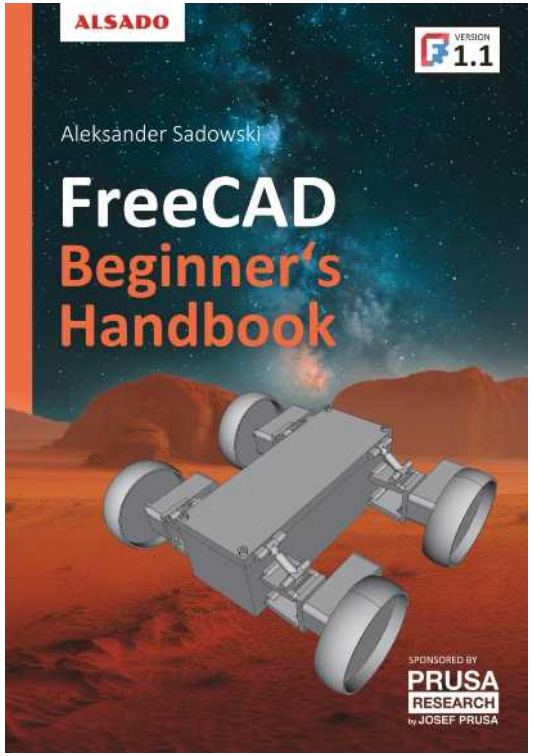
FreeCAD Beginner’s Handbook

- Taking inventors from sketch to 3D printed prototype
- Exploring FreeCAD’s Part Design, Sketcher, and A2Plus workbenches to build and assemble functional models.
- Step-by-step guide with projects and practical examples taken from the industry to reinforce essential techniques and
- Finish off with a practical project, building a Mars Rover in FreeCAD



=> FreeCAD Beginner’s Handbook makes our training available to everyone

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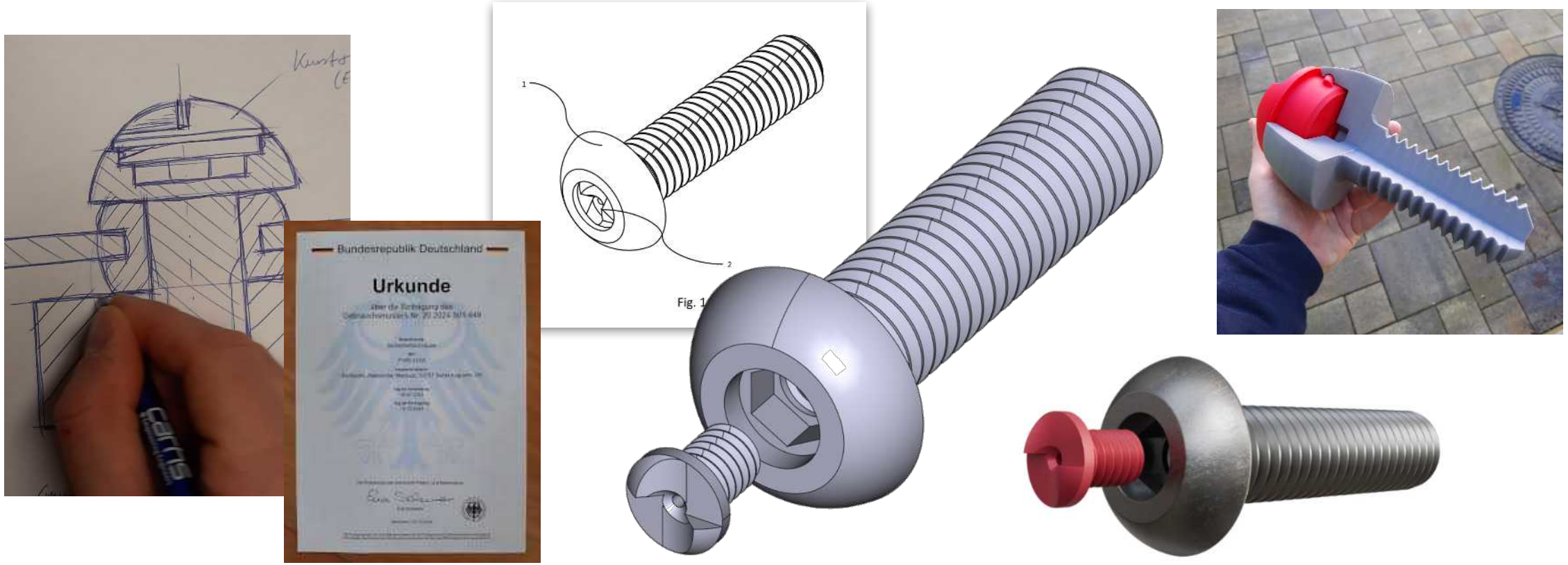
Invention: Security (tamper-resistant) screw



=> Validating the open-source workflow for product development

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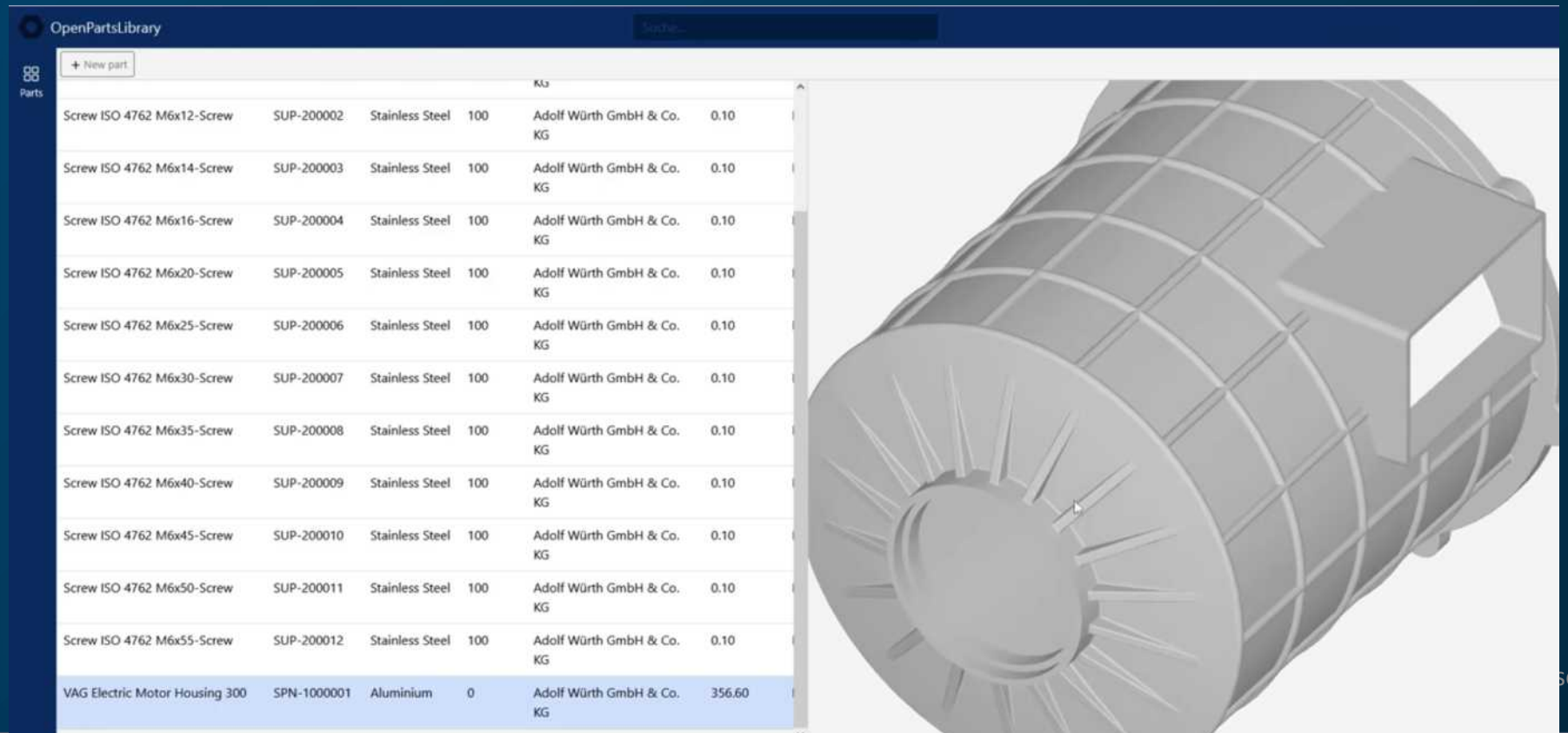
Open-Source PLM/ PDM software

Simplifying the open-source product development process



served.

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Mission: Deutschen Maschinenbau mittels Open-Source-Software wieder auf Vordermann bringen!

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